

CHEATSHEET · QUICK REFERENCE

Lab 31 — OS and Software Troubleshooting

Triage systemd services, fix clock skew and broken dpkg state, use recovery boot modes, control CPU affinity and prevent config drift with Ansible.

Service and hang triage

`systemctl status sshd`
Service state

`systemctl list-dependencies sshd | head -20`
Dependency chain

`systemctl list-units --type=service --state=failed`
Failed units

`strace -p $PID -c -f`
Diagnose a hanging process

Clock skew + broken dpkg

`chronyc tracking`
Check drift (Kerberos fails > 5 min)

`chronyc -a 'burst 4/4'`
Force a sync

`dpkg --configure -a`
Recover a stuck dpkg state

`apt -f install`
Fix broken dependencies

Recovery boot modes

Mode	Linux equivalent
Safe mode	<code>systemd.unit=rescue.target</code>
Single-user	<code>systemd.unit=emergency.target</code>
Reload OS	Reinstall from media
Snapshot	LVM / VM snapshot revert

Affinity, versions, drift

`taskset -cp 0 $PID`
Pin CPU affinity

`renice -n 10 -p $PID`
Lower priority

`apt-mark hold nginx`
Pin a package version

`ansible-playbook playbook.yml`
Idempotent config to prevent drift